

Assignment 1 Automatic Smart Light

Objective: Design a schematic for an automatic light controller that turns a lamp on when the ambient light decreases (dark environment) and turns it off when there is enough light.

Component	Value / Model	Role in Circuit
Photoresistor (LDR)	Depends on ambient light	Senses ambient light
Resistor (Fixed)	10k Ω	Forms voltage divider with LDR
Potentiometer	10k Ω	Allows calibration of sensitivity
NPN Transistor	2N2222	Acts as a switch
Relay	5V coil	Switches high current load (lamp)
Diode	1N4007	Flyback diode for relay
Power Supply	5V DC for logic	Powers the circuit
Connector / Header	2-pin or screw terminal	Lamp connection



How Each Component Works in This Circuit

1. LDR: Detects light. In darkness, resistance is high → voltage at base increases → transistor switches on.
2. Fixed resistor: Completes voltage divider; sets the threshold for light detection.
3. Transistor: Amplifies the small voltage from LDR divider to drive relay current. Acts as an electronic switch.
4. Potentiometer: Adjustable resistor, Set the threshold level (when light turns ON/OFF)
5. Relay: Turns on/off high current lamp without loading the control circuit.
6. Diode: Prevents back EMF from damaging transistor.

Circuit Concept (How it Works)

LDR + resistor + potentiometer → create a voltage divider

Voltage goes to transistor base

When:

Dark → Voltage increases → Transistor ON → Relay ON → Light ON

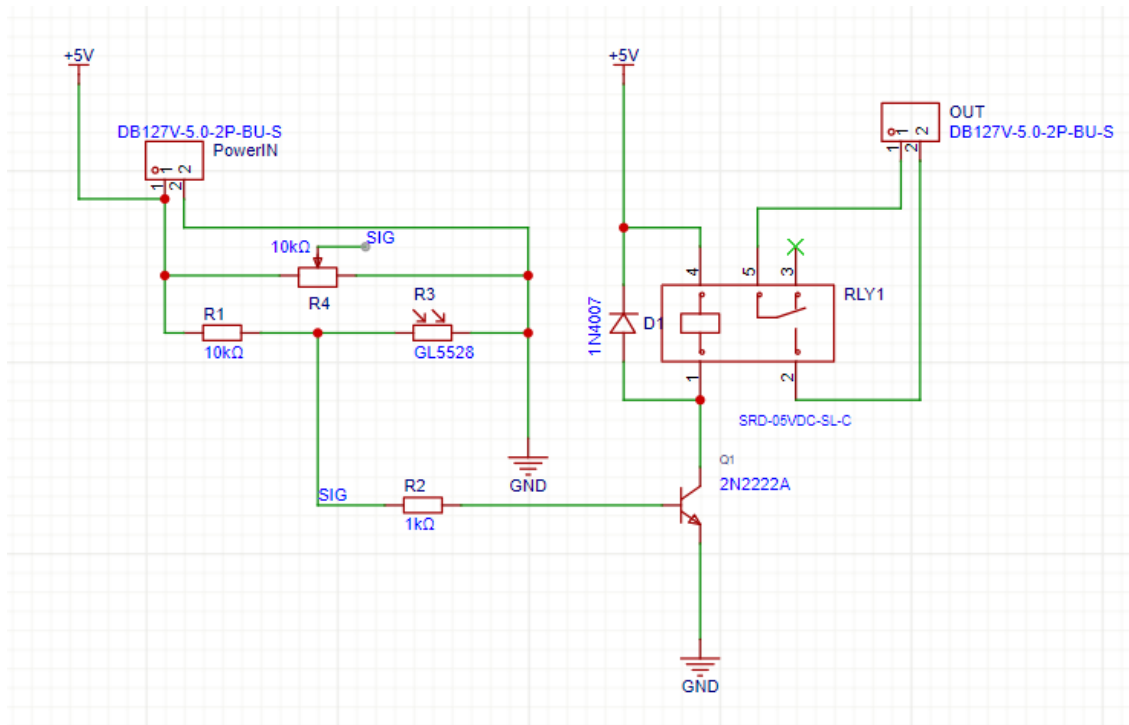
Light → Voltage decreases → Transistor OFF → Relay OFF → Light OFF

Components:

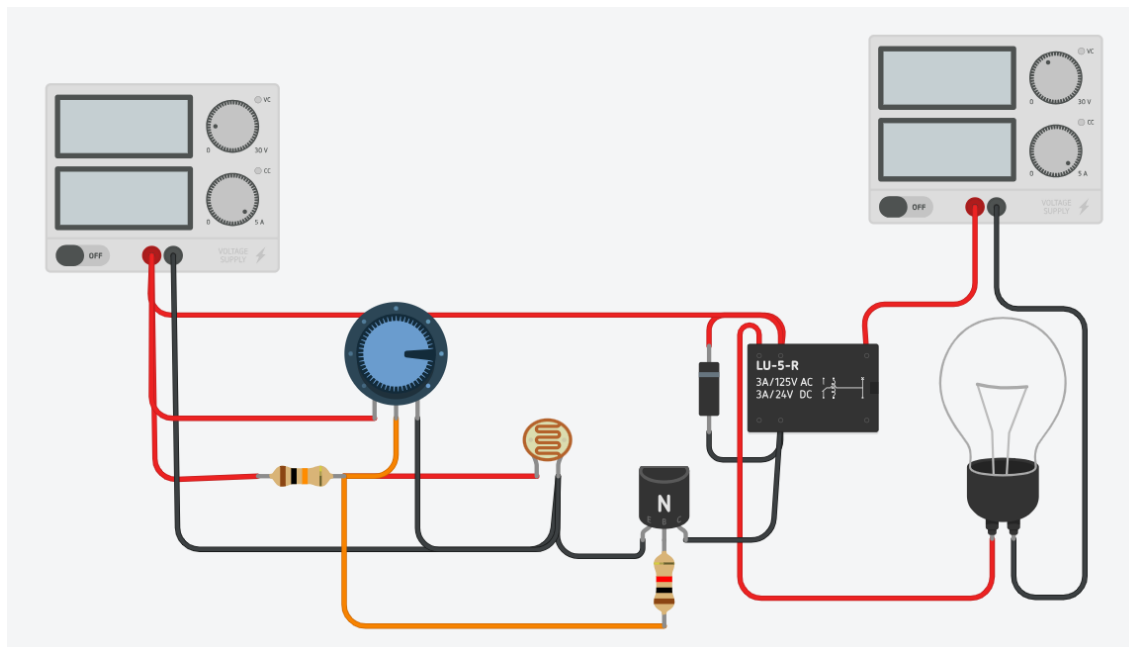
Quantity	Comment	Designator	Supplier Part
1	1N4007	D1	C106903
2	DB127V-5.0-2P-BU-S	OUT,PowerIN	C500019
1	2N2222A	Q1	C118536
1	10kΩ	R1	C119433
1	1kΩ	R2	C173143
1	GL5528	R3	C125627
1	10kΩ	R4	C370978
1	SRD-05VDC-SL-C	RLY1	C35449



Schematic:



Circuit in tinkerCad:



Component description

Transistor (2N2222) Basics



2N2222A	
1	Emitter
2	Base
3	Collector

A transistor is an electronic switch controlled by a small current.

It allows a small signal to control a larger load (like a relay).

Terminal	Symbol	Function
Base	B	Control input
Collector	C	Connected to load (relay)
Emitter	E	Connected to GND

State	Base Voltage	Transistor
OFF	$< 0.7V$	No current flow
ON	$\geq 0.7V$	Current flows

Relay

A relay is an electrically controlled switch.



It allows a low-power circuit (like Arduino or transistor) to control a high-power device (like a lamp)

Relay Terminals

Control Side (Coil)

Switching Side

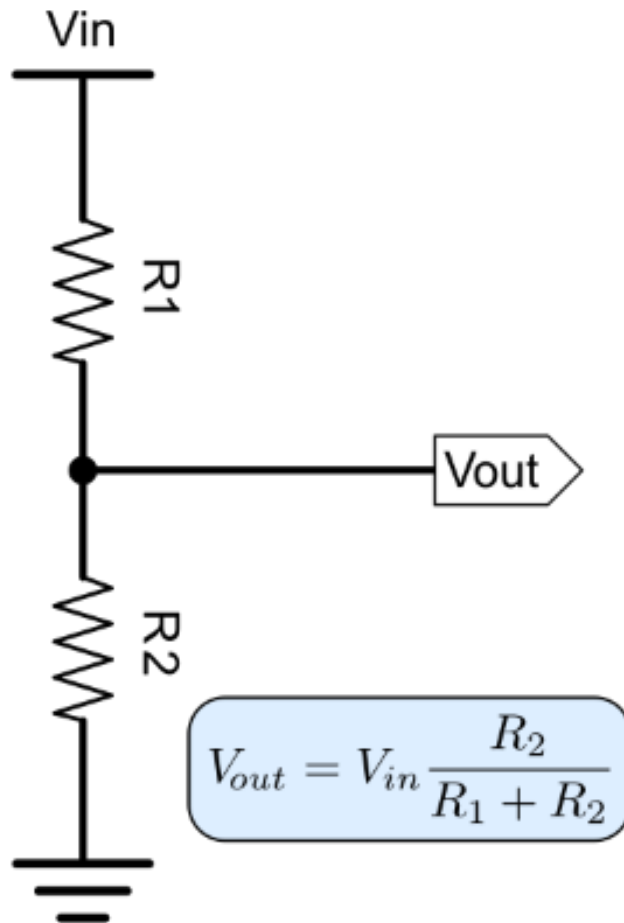
Com: Common , Main connection

NO: Normally open, OFF by default

NC: Normally closed ON by default



Voltage Divider concept



A voltage divider is a circuit that splits voltage using two resistors.

It is used to create a specific voltage level from a source.